

DCent-PredictiveNet: A novel explainable hybrid model for time series forecasting

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ABSTRACT

This work presents a novel hybrid framework called DCEnt-PredictiveNet (deep convolutional neural network (DCNN) + entropy + support vector regressor (SVR)) that concatenate both deep and handcrafted features for time series data analysis and forecasting. From the discrete wavelet transform coefficients of input time series data, computed four different handcrafted entropy features, which were then concatenated with deep features extracted using a modified DCNN. The concatenated deep and handcrafted feature vector was then fed to a SVR for prediction. The DCEnt-PredictiveNet framework was trained and tested on three time series datasets of real-world COVID-19, stock price and traffic information, and achieved mean absolute percentage errors of 0.03 %, 1.53 % and 11.41 % for daily cumulative COVID-19 positive cases, closing stock price, and hourly traffic (vehicle numbers) at one junction predictions, respectively. In addition, we incorporated local interpretable model-agnostic explanations and Shapley additive explanations methods into DCEnt-PredictiveNet to enable visualization of significant features that contributed to the model's decision-making, thereby enhancing its explainability. Our DCEnt-PredictiveNet model yielded promising and interpretable forecasting results, which can facilitate advance resource planning in hospitals for incoming COVID-19 patients, stock market investment planning, and efficient traffic control management.

1. Introduction

Forecasting, the prediction of some future event or events, is significant and challenging problems in majority of the fields which involve time series data. Time series forecasting (TSF) can be short-term (hourly, daily, and weekly, into the future), medium-term (monthly to yearly into the future) and long-term (few yearly into the future) events prediction. The TSF models developed comprises techniques for analyzing time series data to extract significant statistics and other characteristics of the data. These models accurately assess patterns in the data and conveys a statistical correlation between past and present data values. Then the model is used to estimate the future data patterns. Prediction of future events is a important in many types of planning and decision-making process with application to areas such as operations management, marketing, medicine, finance and risk management, economics, industrial process control and demography. The necessity of

time series prediction lies in its wide-ranging applications that require accurate and timely forecasting for optimal decision-making. Due to the wide range of real-world situation that require forecasts, two broad types of forecasting approaches –statistical and Neural Network based methods have been explored and published extensively for developing prediction model. In addition, AI interpretability is crucial for building trust, ensuring compliance, diagnosing errors, and understanding feature importance.

1.1. Related work

Conventionally, statistical, and artificial intelligence (AI) based approaches have been published in literature for developing TSF models in many areas [33,56,74]. Different statistical based methods such as moving average (MA), autoregressive integrated moving average (ARIMA) and exponential smoothening [64,69] are employed

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predominantly for TSF. Even though these methods are efficient to analyze the linear time series data, the nonlinear behavior/patterns of the time series tampers/hinders/limits the use of few statistical methods for developing predictive models. Therefore, to handle the nonlinear patterns of the real-world time series data, the autoregressive conditional heteroskedastic (ARCH), general AR conditional heteroskedastic (GARCH), and threshold AR (TAR) [92] methods have been explored. Still these methods lack in capturing specific nonlinear patterns making it less selected and accurately suited methods for nonlinear TSF task.

To overcome the limitations of statistical models in handling high-dimensional multivariate and non-linear time series, recently, AI-based artificial neural networks (ANNs) [41,49,63,66], Convolutional Neural Network (CNN), Long Short-term Memory (LSTM) [25,71] and genetic algorithms (GAs) have been employed and reported of better forecasting capabilities compared with statistical based methods. ANN models have become more popular in forecasting and have been employed to various time series data for forecasting COVID epidemiological trend [20], electricity price [68], financial distress [13], and flow of data in network traffic analysis [12]. Even though neural network-based models have shown to be advantageous/capable compared to statistical methods in analyzing nonlinear time series, their capability fell short in analyzing time series having combination of both linear and nonlinear patterns [92]; Mehdi and Mehdi [46]. Thus, different hybrid based predictive models have been invented and investigated successfully by combining ANN with different machine learning tools for time series analysis and forecasting in different areas [22,45,58,84,97]. In addition, hybrid, or fusion of statistical methods with neural network or different deep CNNs have been explored by various researchers [21,38,46,5,80] for TSF task.

Thus, in recent years, more hybrid architectures have been developed and applied in many areas with good prediction performance. Hybrid of ARIMA-ANN [21,5,92], ARIMA-support vector regression (SVR)/support vector machine (SVM) [19], ARIMA-MLANN [91], adaptive wavelet transform (AWT) based LSTM-ARIMAX-GARCH [75], exponential smoothing-NN [28], SVR-GA [24], exponential smoothing model (ESM)-ARIMA-BPNN [78], ARIMA- Probabilistic Neural Network (PNN) and ANN-PNN [47], Wavelet- multivariate adaptive regression splines (MARS)-SVR [31], wavelet-adaptive network based fuzzy inference system (ANFIS)- quantum behaved particle swarm optimization (QPSO) [2], Spatio-Temporal Graph Convolutional Network (H-STGCN) (Rui [18]), ARIMA- Nonlinear autoregressive (NAR) [73], ensemble empirical model decomposition (EEMD)-ARMA [16], by combining EEMD with neural network (BPNN or RNN) and SVR [6], two piece scale mixtures of normal (TP-SMN)-ARMA [52], CNN-BiLSTM- efficient channel attention (ECA) [14], CNN-LSTM [27], GA-XGBoost [89], ARIMA+CNN+Seq2Seq LSTM [55,86], multi-head attention + LSTM+SNN with Bayesian Optimization [25], attention+ConvLSTM+two BiLSTM [94], attention + CNN + LSTM [77], ARIMA + LSTM [40], Fast RNN + CNN + BiLSTM [85], CNN + LSTM + Attention mechanism [29], Wavelet + LSTM + adaptive genetic algorithm (AGA) [90], wavelet + gated recurrent unit (GRU) + LSTM [51], Stock generative adversarial network (GAN) with generator (LSTM) + discriminator (CNN) [70], CNN with local/global attention modules + CNN and BiLSTM [61], BiCuda DNN LSTM + CNN [30], temporal CNN with a multi-scale attention mechanism (MA-TCN) + Multiview CNN-BiLSTM-Att [88], emotion enhanced CNN + denoising autoencoder + LSTM [93] have been implemented for the TSF which demonstrated significantly efficient results compared to other (statistical and ANN based) methods in linear and nonlinear time series analysis. However, one hybrid model which is explainable and applicable to different areas for the prediction task has not been explored much. Therefore, in this paper, in comparison to the published time series forecasting models, a novel explainable hybrid model - **DCEnt-PredictiveNet** is proposed and implemented for the time series forecasting task which can be applied to three different areas (medicine, finance and traffic). The proposed hybrid model involves (i) both linear (DWT) and

nonlinear (entropy and CNN) methods to examine the time series data and (ii) the computation of hand-crafted features using entropy and deep features using CNN. Combining these two types of features aims to leverage the strengths of both automated deep learning feature extraction and human-engineered features, resulting in a more comprehensive and robust representation of the data. DCEnt-PredictiveNet is made explainable by employing local model-agnostic interpretable methods (Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP)). Integrating these interpretability methods ensures that the model is not only accurate but also transparent, allowing users to trust and understand its predictions. The motivation behind DCEnt-PredictiveNet is to create a powerful hybrid model that combines the strengths of deep learning and handcrafted features for superior prediction performance while ensuring the model remains interpretable and transparent. This approach addresses both the need for accuracy in complex time series forecasting and the demand for explainable AI. To best of our knowledge, this is the first explainable hybrid model proposed and implemented for three different areas with promising results. Thus, our approach aims to leverage the strengths of hand-crafted and deep features methods to achieve superior prediction accuracy while ensuring model interpretability with the following main contributions.

- (i) Unique hybrid feature combination framework to capture both global and local patterns in the time series data.
- (ii) Use of LIME and SHAP to enhance the transparency of our model to make it easier to understand and trust the model's decision-making process.
- (iii) Rigorous evaluation of our proposed model on three diverse time series datasets – COVID, Stock price and traffic flow.
- (iv) Possibility of scalability and real-time application of our model in future to highlight the practical relevance for dynamic and large-scale environments.

These distinctive features collectively contribute to its outstanding performance and practical relevance, setting it apart from existing hybrid models in the field of time series forecasting.

2. Methodology

In this work, a novel hybrid model called **DCEnt-PredictiveNet** is developed which comprises integration of hand-crafted entropy features with deep features extracted from time series data. Fig. 1 displays the flowchart of forecasting process used to develop time series **DCEnt-PredictiveNet** framework. In hand crafted feature extraction stage, different entropy features are computed from approximate DWT coefficients of time series data, whereas for the deep features extraction stage, deep CNN is used as the backbone framework to extract deep features from time series data. Both the features are concatenated and fed to SVR for prediction. The step-by-step implementation of the DCEnt-PredictiveNet is explained in the following sections.

2.1. Proposed Hybrid Model – DCEnt-PredictiveNet

In this work a novel hybrid framework called **DCEnt-PredictiveNet** is implemented for the TSF. The DCEnt-PredictiveNet framework implemented comprises of Entropy features computation from approximate DWT coefficients of time series data which are concatenated with deep features of time series data extracted using modified DCNN. The concatenated features are then fed to the SVR for performing automated prediction task. The development of DCEnt-PredictiveNet model is explained in the following section.

Fig. 2 shows the proposed explainable hybrid model –DCEnt-PredictiveNet architecture. The proposed explainable DCEnt-PredictiveNet model consists of integration of two stages of feature extraction: deep features and hand-crafted features. In the hand-crafted feature

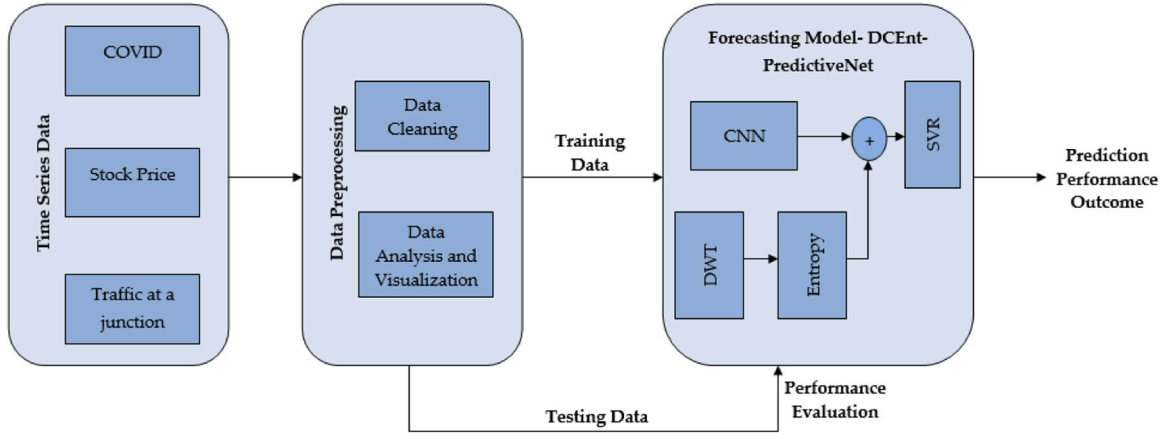


Fig. 1. Flowchart of TSF process for the development of our proposed hybrid model - DCEnt-PredictiveNet.

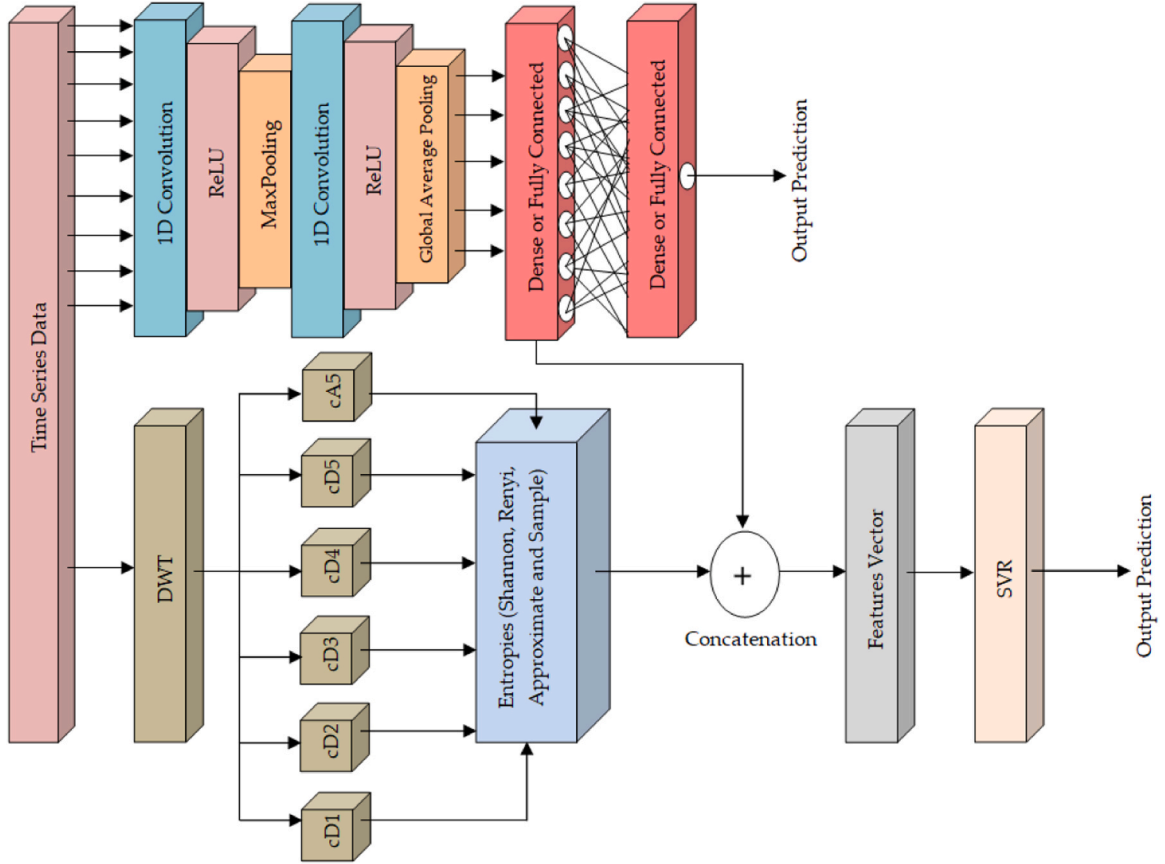


Fig. 2. Proposed DCEnt-PredictiveNet Architecture.

extraction stage, initially the time series data are subjected to DWT to obtain different coefficients. From each approximate DWT coefficients, different entropies such as Shannon entropy, Renyi entropy, Approximation entropy and Sample entropy are computed. Whereas in the deep feature extraction stage, the deep CNN is applied to time series data directly to compute the deep features; features computed from both the stage are concatenated and are fed to SVR for automated prediction task. The architecture is designed to be explainable using two local model-agnostic interpretable methods.

2.1.1. Deep feature extraction stage: deep CNN

Convolutional neural network (CNN) [34] has gained a lot of

popularity in image and signal processing. A typical CNN architecture comprises of convolutional layer, pooling layer and a fully connected layer as three important layers. In this work, 1D CNN is used to analyze the time series data. Initially, time series data of size AAPL – 1006, Covid – 560, Traffic junction1 – 14592, Traffic junction2 – 14592, Traffic junction3 – 14592, Traffic junction4 – 4344 are rearranged for a timestep of 5, depth of 1 and fed as input to CNNs having two blocks of convolutional layers and finally through a dense layer and prediction layer. The features are drawn from final dense layer are concatenated with the entropy of time series data. These combined CNN and entropy features are used to train SVM, and final prediction is obtained.

A 1D CNN model is used get the CNN features from time series data.

As demonstrated in Fig. 2, there are two 1D convolutional layers. First convolution layer uses 64 kernels of height one followed by a rectified linear activation unit (ReLU) and a Maxpooling of size two. First convolution layer uses 16 kernels of height one followed by a ReLU and a global average pooling layer. Lastly, the learned features from previous layers are passed through fully connected (dense) layer with 8 neurons before a linear activated dense layer which makes the prediction. Through numerous trainings, model's hyperparameters values (number of filters, batch size of sample elements, and stride) are finalized in this work.

2.1.2. Hand crafted feature extraction stage: DWT and entropy feature

In order to compute the hand-crafted features set - entropies, the time series data are subjected to five level DWT and from each DWT coefficients computed the different entropy features (Shannon entropy, Approximate Entropy, Renyi Entropy and Sample entropy). The DWT decomposes a time series into different frequency components, allowing the analysis of both time and frequency characteristics simultaneously. This is achieved by applying wavelet filters to the signal at various scales. The time series signal data is first filtered through a filter bank which contains low pass and high pass filters. The low pass filters allow the low frequencies and reject the high-frequency components of the signal and the high pass filter allows the high-frequency component and rejects the low-frequency components. This is Level 1 decomposition and the scale chosen is 2. The output yields low pass (*approximate1 A1*) and high pass (*details1 D1*) sub-bands or coefficients. Approximations represent the general trend or low-frequency components of the time series, whereas the Details capture the fine-scale, high-frequency fluctuations. For the next level of decomposition, the low pass sub-band (*A1*) is again filtered to yield Level 2 low pass (*approximate2, A2*) and high pass (*details2 D2*) sub-bands. The above procedure is performed until the number of samples in the signal is less than the length of the wavelet filter applied and hence the step of maximum decomposition is reached. In this work, the Db4 wavelet of level 5 is used to extract high- and low-level frequency coefficients from the training dataset. From D_1 , D_2 , D_3 , D_4 , D_5 , and A_5 , different entropies are computed as hand-crafted features. Entropy measures the uncertainty or randomness in the data [1,72]. In the context of DWT coefficients, entropy-based features quantify the complexity and information content at different scales of the time series. In our work, the following entropies are calculated.

Shannon Entropy (ShEn) measures the average uncertainty in a set of possible outcomes, for wavelet coefficients, it quantifies the information content. It is computed using,

$$ShEn = - \sum_i p_i \log p_i$$

Where p_i are the probabilities associated with the wavelet coefficients.

Renyi Entropy (ReEn) is a generalization of ShEn and is used to extract valuable insights into the complexity and information content of different frequency components of the time series data. It is defined as,

$$ReEn = \frac{1}{1-\alpha} \log \left(\sum_i p_i^\alpha \right)$$

Where α is a parameter that determines the order of the entropy.

Approximation entropy (ApEn) is used to quantify the amount of regularity and complexity in the time series signal at different scales. It is computed using,

$$ApEn = \phi_m(r) - \phi_{m+1}(r)$$

Where r is the similarity criterion, typically a fraction of the standard deviation of the time series, $\phi_m(r)$ is the average natural logarithm of the fraction of pairs of patterns of length m that are within r of each other.

Sample entropy (SampEn) used to capture the regularity and unpredictability/complexity of the wavelet coefficients (time series at

different scales).

$$SampEn = - \log \frac{A}{B}$$

Where A is the number of pairs of sequences of length $m+1$ and B is the number of pairs of sequences of length m .

Each entropy measure captures different aspects of the time series' dynamics, which can significantly enhance the feature representation used for prediction models. Different entropy measures provide unique advantages when applied to time series data - The *ShEn* provides a fundamental measure of uncertainty and information content, *ReEn* offers flexibility and adaptability by varying the parameter α , capturing different aspects of the data distribution, *SampEn* reduces bias, robust to noise, and effective for shorter time series, *ApEn* computationally simpler, historically significant, and effective in detecting regularity changes. Thus, by computing and incorporating different entropy measures like *ReEn*, *SampEn*, *ApEn* and *ShEn*, the DCEnt-PredictiveNet model leverages a comprehensive set of features that capture the intricate patterns and complexities in time series data. This approach enhances the model's predictive performance and interpretability, addressing the critical need for accurate and explainable time series forecasting.

2.1.3. Concatenation and prediction based on SVR

The hand-crafted entropy features obtained are concatenated with deep activated CNN features and fed to Support Vector Regressor (SVR) for the prediction task. The SVR is an advanced machine learning algorithm used to solve regression problems. It performs exceptionally well in problems that are related to higher dimensional spaces and can achieve high efficiency in regression by constructing a decision boundary or hyper plane between non-identical data points along with support vectors on both sides of the boundary.

2.2. Explainable DCEnt-PredictiveNet

The proposed DCEnt-PredictiveNet is made explainable by employing local model-agnostic interpretable methods. Two local model-agnostic methods such as LIME and SHAP are used to explain the proposed hybrid model's predictions. In addition, the contribution of each feature together with information on the features which have significant contribution in prediction are obtained.

2.2.1. LIME

In this work, LIME explanation technique originally developed by Ribeiro et al. [60] is used to make the hybrid model's decision interpretable. This method used explains the reason behind the model's prediction decision for a data sample by finding important predictors and fitting a simple interpretable model. LIME algorithm involves creating a new dataset consisting of perturbed data samples, predicting corresponding predictions on the new dataset, calculating weight for perturbed data based on the Euclidean distance of perturbed data to the original observation, creating a linear model on perturbed data using weights and then explaining the prediction by new linear model. The LIME gives a list of explanations, reflecting the contribution of each feature to the prediction of a data sample. **SHAP** by Lundberg and Lee [42] is a mathematical method to explain the predictions of machine learning models by fairly distributing both gains and costs to features working in a coalition. It is based on the Shapley values, a widely used method in game theory. The Shapley value is the average marginal contribution of a feature value across all possible coalitions. SHAP values can be used to determine the most important features and their influence on the model's prediction. SHAP values inform the amount by which each feature contributed to the model's prediction. Large positive/negative SHAP values indicate that the feature had a significant impact on the model's prediction.

3. Experiment results

This section presents the extensive experiments conducted to validate our DCEnt-PredictiveNet using three different datasets with the prediction results obtained. In addition, demonstrates the results of interpretability to explain our proposed model.

3.1. Time series data

In this work, three different time series data were obtained for validation of proposed DCEnt-PredictiveNet model. Dataset 1: time series data on COVID in Singapore from data world, Dataset 2: Stock prices data from Apple Inc (AAPL) database and Dataset 3: the *Traffic Prediction Dataset* obtained from Kaggle contained the 48120 observations of the number of vehicles each hour in four different junctions. Time in hourly frequency is used as independent variables whereas the number of vehicles in that hour is used as dependent variables.

3.1.1. Data analysis and visualization

In order to recognize and interpret the patterns present in the time series data, visualization of daily closing stock price, daily cumulative COVID cases and hourly number of vehicles (traffic) at one junction are performed using Microsoft Power Business Intelligence (BI) tool. Fig. 3 shows the visualization of multiple time series data (dataset 1, dataset 2

and dataset 3) using autocorrelation function (ACF). Fig. 3 shows the ACF for daily cumulative COVID cases, with 5 % significance limits, daily closing stock price, with 5 % significance limits, and hourly traffic (number of vehicles) at one junction. For a stationary time series data, a sinusoidal ACF pattern – the rate of decrease or decay in ACF values to 0 followed by sinusoidal pattern, is observed. ACF is very persistent; that is, it decays very slowly and exhibits sample autocorrelations that are still rather large even at long lags. This behavior is characteristic of a nonstationary time series.

3.1.2. Implementation and hyper parameter tuning

We implemented our 1D CNN architectures with keras 2.4.3 and SVM with scikit-learn 0.24.1 on Python 3.8.8. All the experiments are run on NVIDIA GeForce GTX1650. In 1D CNN training, the batch size is 32 and the learning rate is initially set to 0.0001 and is decayed by 1 % patience 3 epochs. The optimizer is the Adam. In this work, through multiple training, model's hyperparameters values (number of filters and stride) are selected. For the other methods, the performances are obtained with their specific settings.

The performance of the SVM is hyperparameter tuned for best possible results. We set the hyper-parameter search spaces and search algorithm of GridSearchCV is performed. The model is further trained on both the training and validation data with the selected hyperparameters, and then are validated on the corresponding test sets. The

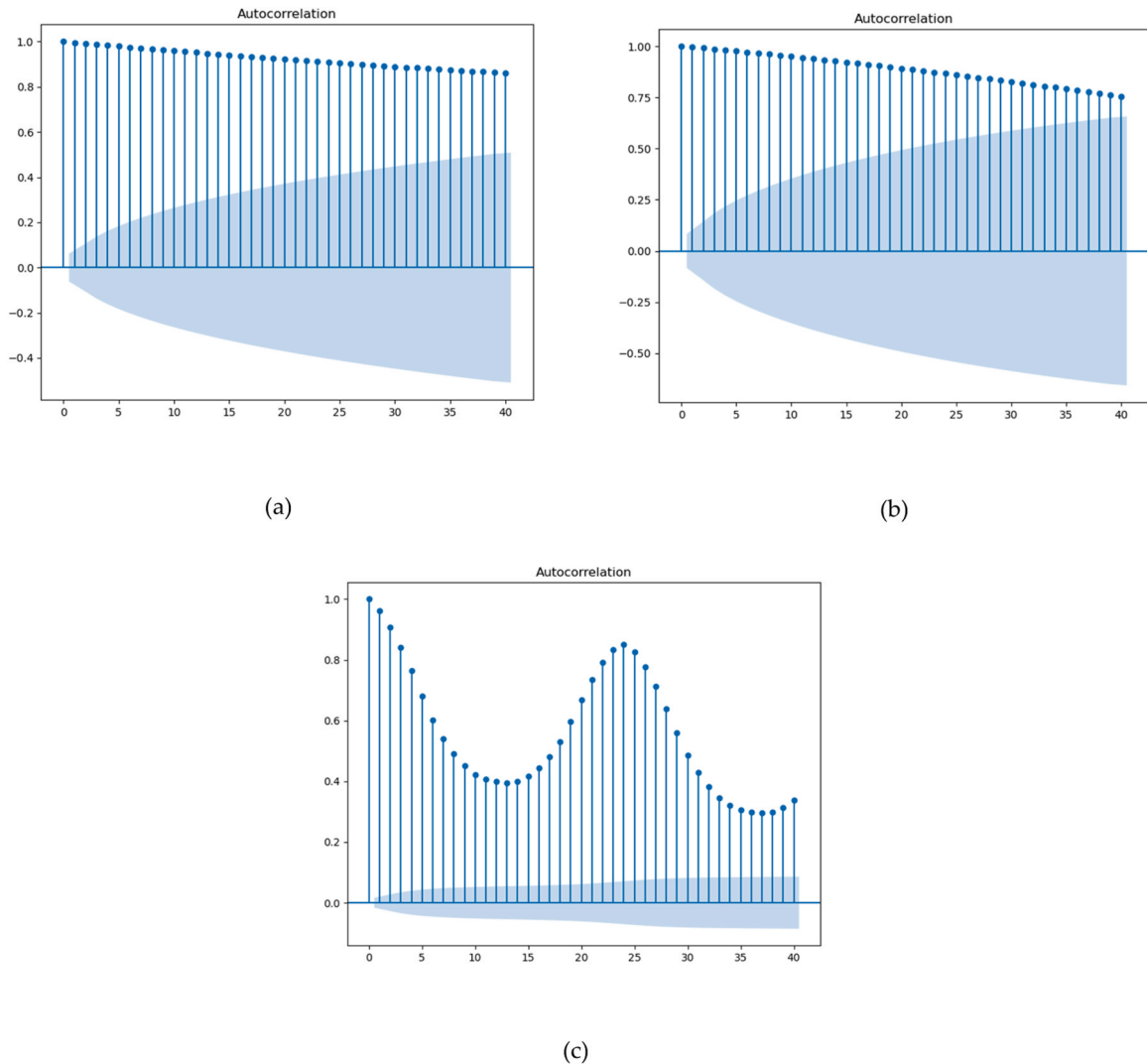


Fig. 3. The ACF for (a) Stock Price, (b) COVID, (c) Traffic (number of vehicles) at junction 1.

predictions are calculated for the test features. The Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) metrics are used to check the efficiency of the model.

3.2. Results of prediction

The prediction performance of our DCent-PredictiveNet for three different applications are shown in Table 1, in addition the actual and predicted results in Table 2 and Fig. 4. It is evident from the Table 1 that DCent-PredictiveNet achieved best performance in prediction of daily cumulative COVID positive cases with MAPE of 0.03 % and MSE of 482.02. The same model demonstrated reasonably good performance in prediction of daily closing stock price and hourly traffic at one junction with MAPE of 1.53 % and 11.41 % respectively.

3.2.1. Ablation study

An ablation study is performed to identify the individual and / or combined impact of deep features and hand-crafted entropy features on overall performance of our hybrid model. Table 3 demonstrates the results of ablation study conducted using stock price and COVID datasets. Through this experiment, analyzed the effect of (i) only the deep features extracted using CNN, (ii) only the hand-crafted four different entropy features (individual and / or combinations), and (iii) concatenation of hand-crafted entropy features with deep features, in the performance of DCent-PredictiveNet model.

It is evident from the Table 3 that, deep features or hand-crafted features if used individually, degraded the performance of DCent-PredictiveNet with higher MSE and MAPE. Concatenating hand-crafted features with deep features enhanced the performance of our model with least MSE and MAPE in both predictions (COVID and stock price)."

3.3. Results of interpretability

In this work, two local model-agnostic methods, LIME and SHAP, are implemented to explain the predictions of DCent-PredictiveNet. Figs. 5–7 show the visual illustration of SHAP, and LIME results obtained for prediction of daily cumulative COVID positive cases, daily closing stock price and hourly traffic at one junction.

From Fig. 5, it is shown that at 0th instance the predicted number of cumulative daily COVID positive cases are 60871.16 for which both the deep (5 features) and hand-crafted (Shannon, approximate and sample entropies from different DWT coefficients) features have contributed positively. At 5th and 10th instances most of the deep features exhibited positive contribution while hand-crafted features shown both positive and negative influence on the predicted number of cumulative daily COVID positive cases. The SHAP waterfall plots (Fig. 5) aid to visualize the Shapely values for each of the features. It is exhibited in this plot the influence of each feature on amount by which the predicted COVID positive case number increases or decreases. At 0th instance (Fig. 5), the base value or average predicted number of COVID positive cases $E[f(x)]$ is 41392.66 and the ending value or predicted number of COVID positive

Table 1

Results of DCent-PredictiveNet for prediction of cumulative daily COVID cases, daily closing stock price and hourly traffic at one junction.

Prediction	Model	MSE	MAPE
Closing stock price	CNN	13.33	7.23 %
	DCent-PredictiveNet	6.03	1.53 %
Hourly Traffic (vehicle numbers) at one junction	CNN	724.51	38.62 %
	DCent-PredictiveNet	87.39	11.41 %
Cumulative daily COVID cases	CNN	18534.14	1.73 %
	DCent-PredictiveNet	482.02	0.03 %

Table 2

Actual and predicted results for three different time series datasets taken at 0th, 5th and 10th instances.

Time Series Data	Instances	Actual	Predicted
COVID	0	60851	60871.16
	5	60966	60985.83
	10	61121	61131.41
Stock Price	0	121.19	120.95
	5	116.87	118.09
	10	111.2	117.23
Hourly traffic (number of vehicles) at one junction	0	98.48	100.98
	5	78.96	94.71
	10	109.30	110.06

cases $f(x)$ is 60871.16. Most of the deep features have contributed to increasing the predicted number of COVID positive cases while hand-crafted features have shown their influence in both increasing and decreasing the predicted number of COVID positive cases. The other waterfall plots obtained for 5th and 10th instances exhibits the similar observation for COVID dataset.

From Fig. 6 LIME results, it is shown that at 0th, 5th and 10th instances most of the hand-crafted features have contributed positively and most of the deep features exhibited negative influence on the predicted number of cumulative daily stock price prediction. From waterfall plots, at 0th, 5th and 10th instances (Fig. 6), most of the hand-crafted features have contributed to increasing the stock price prediction while deep features have shown their influence in decreasing the stock price prediction.

In the prediction of daily closing stock price and hourly traffic (number of vehicles at junction), according to both LIME and SHAP (Fig. 7) plots, hand-crafted features exhibited significant influence on DCent-PredictiveNet's decision-making compared to deep features for three different instances (0th, 5th and 10th). Thus, from both SHAP and LIME plots it is evident that deep features showed significant impact compared to hand-crafted on DCent-PredictiveNet's COVID prediction task.

4. Discussion

In recent years, more hybrid architectures have been proposed and applied in many areas with good prediction performance. In this paper, focused mainly on TSF application (i) to medicine – prediction of COVID, (ii) finance – prediction of stock price and (iii) traffic flow – prediction of number of vehicles at junction. Summary of recent various hybrid models published for the TSF in three different areas (medicine, finance, and traffic flow) are tabulated in Table 4.

Time Series Forecasting in Medicine – Recent hybrid models for COVID prediction

[23] proposed a hybrid model that incorporated EEMD and ANN for predicting the COVID epidemic with MSE of 3.76E-05. Castillo and Melin [10] presented a hybrid approach by combining fractal theory and fuzzy logic which reported an error of 0.02 % for forecasting COVID confirmed and death cases. Alakus and Turkoglu [3] proposed a clinical predictive model for laboratory data analysis to forecast possible COVID cases using CNN-LSTM and CNN-RNN hybrid methods. Their study reported the predictive performance of 84.16 % accuracy by CNN-LSTM and 85.66 % accuracy by CNN-RNN. In order to forecast the one-month COVID casualties, [67] proposed a hybrid model which involves the DWT combined with ARIMA and achieved the MAPE of 1.248 % (Spain dataset), 5.623 % (UK dataset), 15.625 % (USA dataset), 18.533 % (France dataset) and 2.804 (Italy dataset). Bhardwaj and Bangia [7] demonstrated Mean Absolute Scaled Error (MASE) values of 0.06 (China data), 5.76 (India data), 3.39 (America data) for confirmed, 0.99 (China data), 5.52 (India data), 1.34 (America data) for death and 0.15 (China data), 3.11 (India data), 5.57 (America data) for recovered COVID daily cases prediction using their

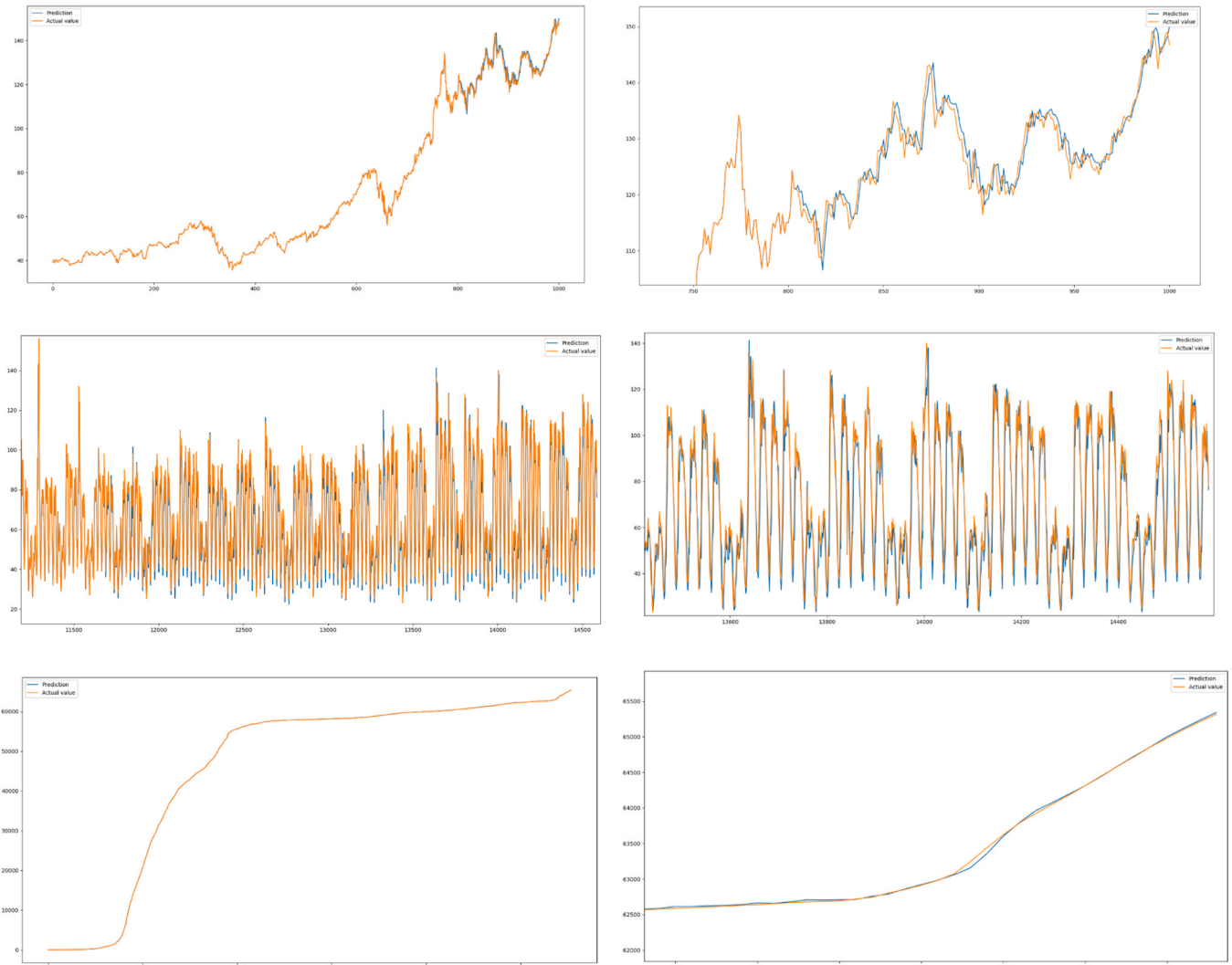


Fig. 4. Plot of DCent-PredictiveNet showing the actual and predicted values for daily closing stock price (topmost), hourly traffic (center) and daily COVID positive cases (bottom one). X-axis: samples or instances, Y-axis: actual and predicted values.

Table 3

Results of ablation study.

Model	Deep Features	Hand crafted features computed from DWT Coefficients				MSE	MAPE
		Shannon	Renyi	Approximate	Sample		
DCent-PredictiveNet (Stock price)	✓	✓	✓	✓	✓	6.03	1.53
	X	✓	✓	✓	X	6.84	1.62
	X	✓	X	✓	✓	7.10	1.61
	X	✓	✓	✓	✓	11.50	2.09
	X	X	✓	✓	X	22.86	2.80
	X	X	✓	X	✓	21.68	2.81
	X	X	✓	X	X	21.09	2.82
	✓	X	X	X	X	30.63	3.19
DCent-PredictiveNet (COVID)	✓	✓	✓	✓	✓	482.02	0.03
	X	✓	✓	✓	✓	20764.52	0.12
	X	X	✓	✓	✓	439825.85	0.88
	X	X	✓	✓	X	433294.95	0.88
	X	X	✓	X	X	415849.29	0.89
	X	X	✓	X	✓	461081.55	0.92
	✓	X	X	X	X	2653446.98	2.27

Wavelet-Neuronal-Fuzzification hybrid model.

In order to improve the COVID cases prediction with R^2 score of 0.9763, [50] implemented a hybrid model using Cauchy exploration strategy Beetle Antennae Search (CEBAS)-ANFIS. A novel forecasting

framework, Optimized data assimilated NN (ODANN) which combines natural language processing (NLP) for producing features from Twitter data (written in English) as input to deep NN and assimilating with historical time-series data for forecasting the global growth rate in the

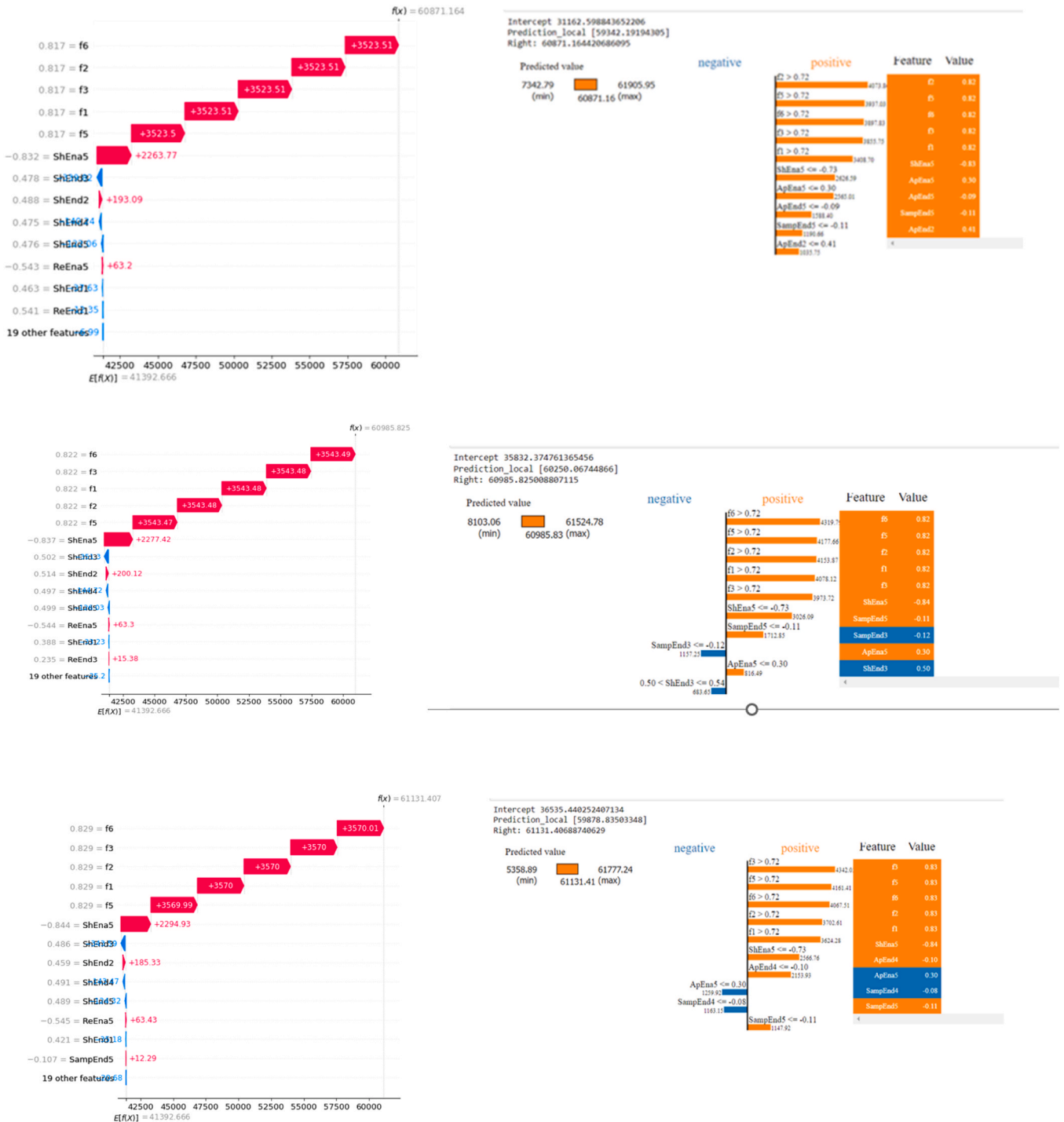


Fig. 5. Visualization of features contribution using SHAP waterfall plot (left) and LIME (right) in the daily COVID prediction.

total number of confirmed COVID cases on daily basis – termed as G parameter, is proposed by Alvin et al. [4]. They demonstrated lowest MSE of 0.0000130, MAE of 0.002972 for 3 days in advance COVID cases prediction using ODANN+ data assimilation component + Skip-gram NLP feature extraction method, whereas using ODANN+ data assimilation component + Word2Vec – Continuous Bag of Words (CBOW) NLP feature extraction method, their study reported MSE of 0.0000080 for 5 days in advance COVID cases prediction. Shoaib et al., 2021 proposed a Susceptible-Infectious-Recovered (SIR) model for COVID dynamics prediction using hybrid computing technique: NAR with RBFs neural network which showed the lowest MSE. A two-layer nested

heterogeneous ensemble learning-based forecasting approach is implemented by Cui et al. [17] that merges linear regression (LR), SVM, and extreme learning machine (ELM) to forecast the COVID-19 mortality trends in various countries. Prasanth et al., [59] developed a hybrid GWO-LSTM framework, where the parameters of LSTM network are optimized using Grey Wolf Optimizer (GWO), which reported a MAPE of 3.452 for total cumulative cases, 7.140 for daily new cases and 0.304 for total cumulative deaths prediction due to COVID. Kumar et al. [32] proposed a novel hybrid fuzzy time series model (ANHFTS) based on Modified Fuzzy c-means (MFCM), for the prediction of upcoming (31 days in advance forecasting) COVID-19 infected cases and deaths in

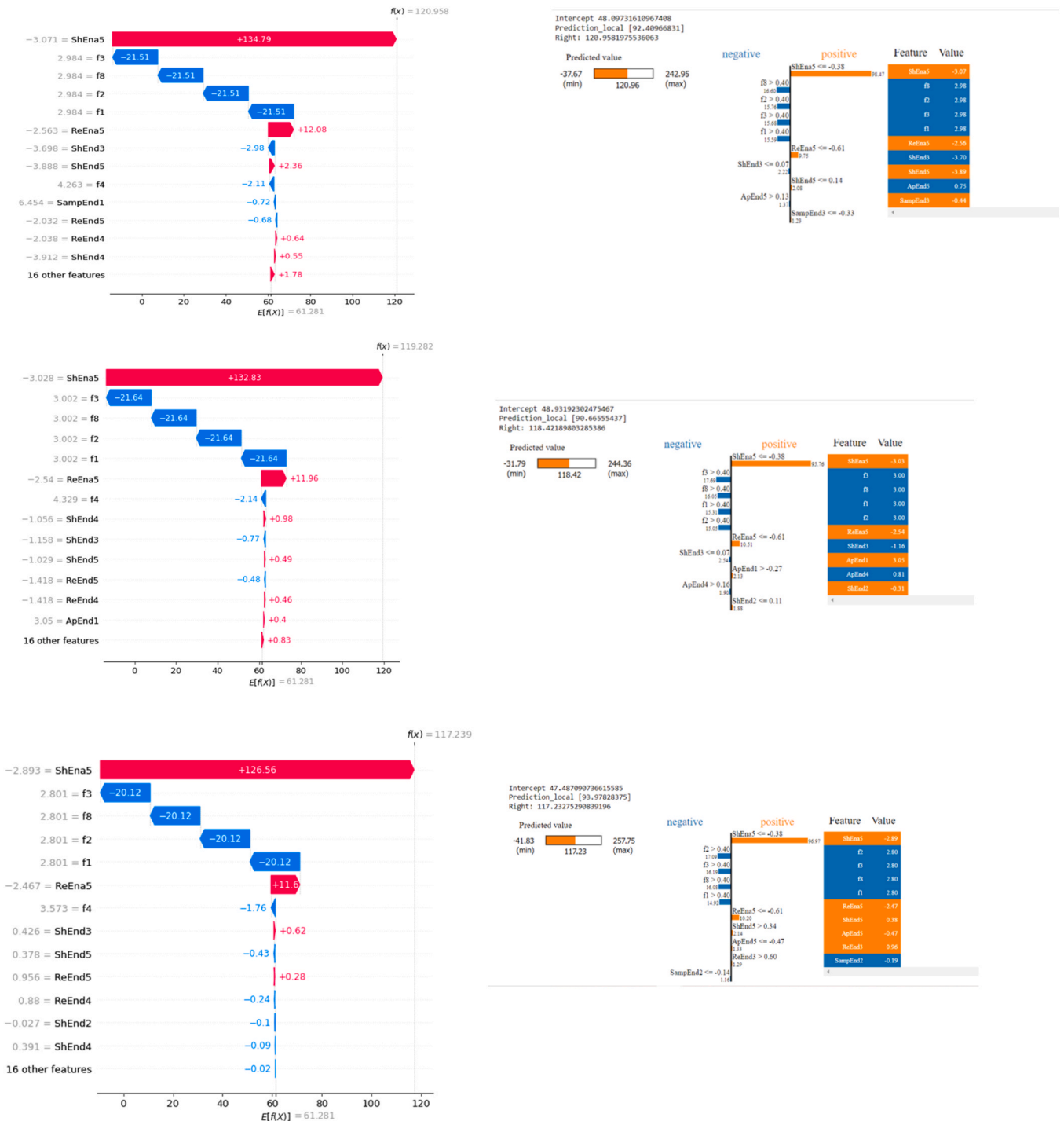


Fig. 6. Visualization of features contribution using SHAP waterfall plot (left) and LIME (right) in the daily closing stock price prediction.

India. Their experiment claimed MSE of 191360.0033 and RMSE of 427.4471 for one-month COVID infected cases forecasting, MSE of 126.2271 and RMSE of 11.2351 for one-month COVID deaths forecasting.

Time Series Forecasting in Finance – Recent hybrid models for Stock Price forecasting/Prediction

From past two decades, various hybrid models such as ARIMA+ANN [92], ARIMA+PNN [47], wavelet + ANFIS with QPSO tuning [2] for exchange rate forecasting, and ARIMA+ANN for three-step ahead stock price prediction [5], and ensemble of EMD and CEEMDAN with LSTM for one-step-ahead forecasting [9]. ESM + ARIMA + BPNN [78] to predict the monthly trend of the stock price index, WT+ MARS, and SVR

(called Wavelet-MARS-SVR) [31], ARIMA+SVM [57], EEMD with neural network (BPNN or RNN) and SVR [6], temporal convolutional network (TCN)-Stock2Vec model [79], and LSTM+EEMD [87] for stock price prediction/forecasting. Recently, [83] proposed a new method by integrating Doc2Vec + Stacked auto-encoder + wavelet transform + LSTM for stock price prediction/forecasting. They used Doc2Vec to create input text features, stacked auto-encoder to reduce the dimensionality of input features data, wavelet transform to decompose the input data and LSTM to predict the stock price. Their proposed model showed prediction results MAE of 0.019 and RMSE of 0.110. [76] developed multi-scale nonlinear ensemble of optimal feature computation (variational mode decomposition (VMD) and auto-encoder) with

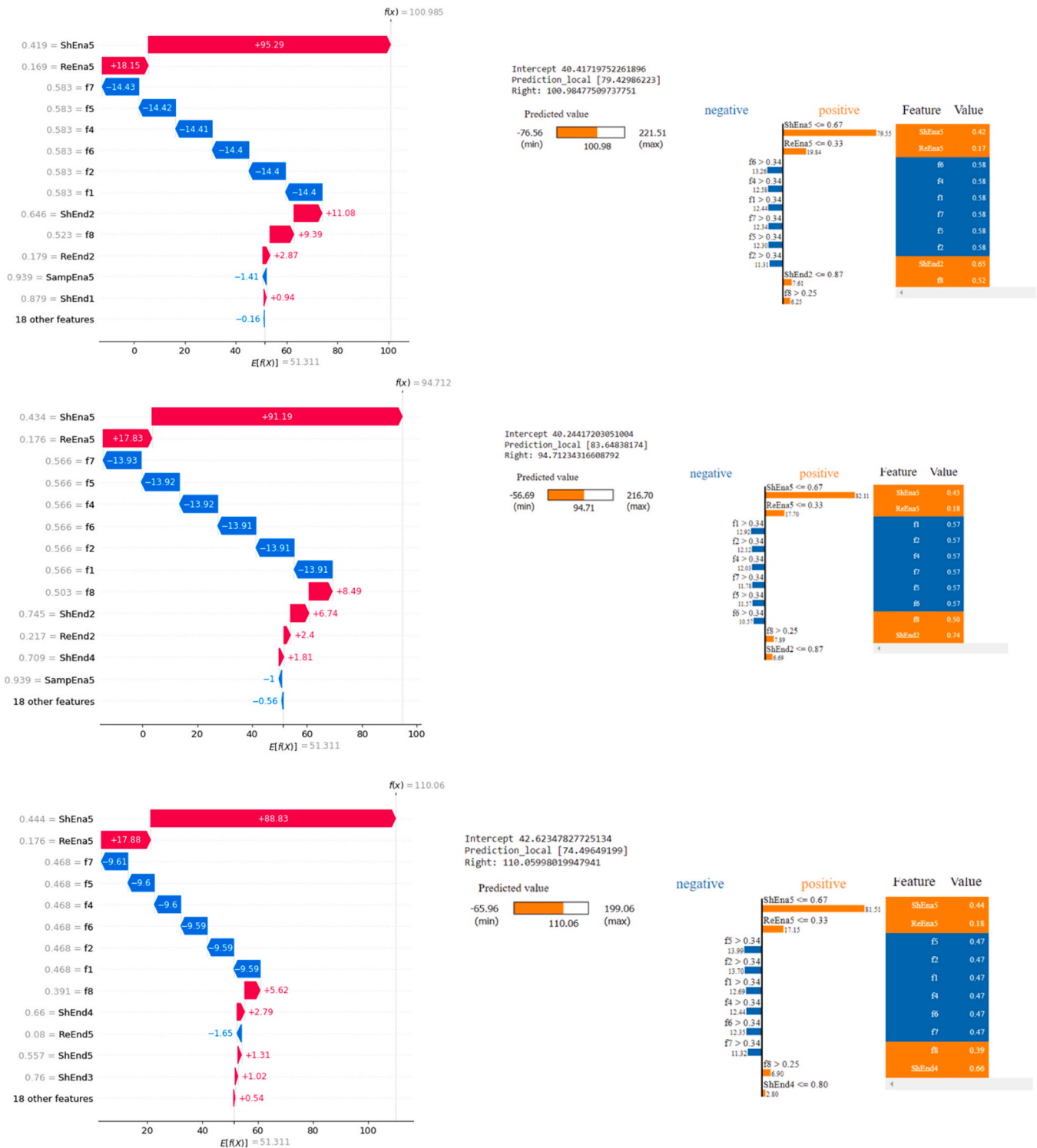


Fig. 7. Visualization of features contribution using SHAP waterfall plot (left) and LIME (right) in the hourly traffic prediction.

two-stage deep neural network (RNN and LSTM) and Gaussian process regression for stock index prediction and uncertainty analysis. Their model demonstrated MAPE of 0.55 % (S&P 500 data), 0.65 % (Dow Jones index data) and 1.11 % (MASDAQ data). Another prediction model combining CNN-BiLSTM and ECA is implemented by [14] which demonstrated effectiveness and feasibility compared to other existing methods with best MAE of 0.103. [27] implemented a hybrid framework by combining CNN+LSTM, where CNN is used to classify the investors' hidden sentiments and LSTM to predict the stock price using the results

of CNN (sentiment analysis) and other technical indicators as input predictors. Their experiment conducted in six different industries showed an average MAPE of 0.0224 for the prediction of stock. [89] proposed a hybrid model called GA-XGBoost comprising of feature set expansion, data organization and optimal feature selection demonstrated an accuracy of 93.82 % in prediction of stock price. To predict the trend of stock price movement by analyzing both linear and non-linear time-series components, [86], developed Residual-CNN-Seq2Seq (RCSNet) hybrid model. Their framework

Table 4

Summary of recently published AI-based hybrid time series forecasting/predictive models.

Author (Year)	Application	Data	Methods	Results
Stock Price				
Wang et al. [79]	To predict the stock closing price	S&P 500 – next day’s closing price as target	TCN-Stock2Vec model	MAPE: 0.61 %
Yang et al. [87]	Prediction of future stock prices	Stock indices data from Yahoo Finance. Four stock indices: ASX, DAX, HSI and SP500	LSTM-EEMD	RMSE: 0.324 (DAX) and 0.101 (ASX) MAE: 0.217 (DAX) and 0.045 (ASX)
Bashar and Maysam [6]	To predict stock market index	S&P 500 daily closing price Nikkie 225 index – daily closing price Daily closing price for the FTSE100 index from the London stock market	EEMD-GA-WA EEMD + neural network (BPNN/RNN) and SVR + weighted ensemble average using a GA to optimize	MSE: 518.90 (FTSE100 data), 118.46 (S&P500), 7254 (Nikkei225)
Ji Xuan et al. [83]	to forecast the stock price	comments from the investors and companies’ news daily transaction data of “Meinian Health” from the Tushare financial database	Integration of Doc2Vec + SAE + WT + LSTM	Doc-W-LSTM, MAE: 0.019, RMSE: 0.110
Chen Y et al. [14]	To predict the stock price	Shanghai Composite Index, China Unicom, and CSI 300	CNN-BiLSTM-ECA	Best MAE: 39.111 (CSI 300), 0.103 (China Unicom), 28.349 (SCI) for 10 time steps prediction
Jing N et al. [27]	To predict stock price	Shanghai Stock Exchange	CNN (sentiment analysis/classification) + LSTM (prediction)	Average MAPE = 0.0224
Yun KK et al. [89]	Prediction of stock price direction	The Korea composite stock price index 200 (KOSPI) to predict the stock index’s up or down movement direction on the next day	Hybrid GA-XGBoost	Accuracy = 93.82 %
Yang Z et al. [86]	To predict stock price movement trend	S&P 500 index stock dataset	Residual-CNN-Seq2Seq (RCSNet): which integrates ARIMA + CNN + sequence-to-sequence LSTM	MAE = 39.456 (one day ahead), 53.176 (3 days ahead), 80.777 (7 days ahead) and 129.269 (14 days ahead)
Nti IK et al. [55]	Stock market prediction	tweets, Web news, forum discussion, Google trends, historical macroeconomic variables, and past stock data from Ghana stock exchange	CNN+LSTM named IK-ConvLSTM	Prediction accuracy = 98.31 % Sensitivity = 89.39 % Specificity = 99.75 %
Yadav K et al. [85]	To predict abrupt changes in the stock prices of a company	Stock values of 4 companies: Apple, Facebook, Nike, and Uber	Fast RNN+ CNN + Bi-LSTM	RMSE = 0.205647 (Apple stock values) RMSE = 0.15137204 (Facebook stock values) RMSE = 0.039458341 (Nike stock values) RMSE = 0.0218099 (Uber stock values)
Kamara AF et al. [29]	For forecasting stock prices	Johnson & Johnson NYSE	CNN (captures short-term stock signals) + LSTM (captures long-term stock signals) + Attention mechanism (captures signals from too long stock price data) Also called attention-based CNN + Contextual Bidirectional LSTM	MAPE: 0.921
Zeng X et al. [90]	Forecasting stock index	Dow Jones Industrial Average (DJIA) – NYSE, S&P 500, Nikkei 225 of Tokyo, Hang Seng Index of Hong Kong Market, CSI300 of Chinese mainland stock market, NIFTY50 index of India	WT + LSTM + AGA	MAPE = 6.245 (DJIA data), 5.683 (S&P 500 data), 25.084 (HangSeng data), 16.772 (Nikkei data), 30.603 (CSI300 data), 9.289 (Nifty50 data)
Subba Rao P et al. [70]	For stock market prices prediction	National Stock Exchange dataset	Stock-GAN with generator (LSTM) and discriminator (CNN) To improve the performance of Stock-GAN proposed – GAN -HPA	Best MSE of GAN-HPA: 0.000021
Ronaghi F et al. [61]	To predict stock price movement	COVID19 PROMO with effects of internet-based and social media trends	COVID–19 HPSMP framework: 1st path -CNN with local/global attention modules + 2nd parallel path -CNN and Bidirectional LSTM	Accuracy = 66.48 % Sensitivity = 60 % Specificity = 71.11 %
Kanwal A et al. [30]	Prediction of stock prices (stock market performance index)	Two datasets: DAX performance Index of Frankfurt Stock Exchange, HIS of Hong Kong Stock Exchange	Bidirectional Cuda DNN LSTM + 1d CNN	MAE: 188.279
Yuan C et al. [88]	Stock market forecasting	China’s Market Sentiment Index (NMSI), COVID–19 Index and CSI 300 Index	COVID–19-MLSF: MA-TCN + MVCNN-BiLSTM-Att	Accuracy = 58.85 %
Zhao Y et al. [93]	For stock price movement prediction	HIS of Hong Kong Stock Exchange	Sentimental analysis DLSTM (SA-DLSTM): Emotion enhanced CNN (ECNN) (sentiment analysis) + denoising autoencoder (DAE) + LSTM	Average MAPE = 1.10
Traffic Flow				
Cao et al. [8]	Traffic forecasting	Taxi GPS dataset (traffic speed and volume) from Chengdu China	Flow and wavelet-integrated spatio-temporal network (FW-STN) with CNN+LSTM	MAPE = 16.83 %
Chen et al. [15]	To predict traffic flow under different time intervals ahead.	traffic flow data from loop detectors installed in freeways in Minnesota State, USA	EEMD + ANN	MAPE = 9.554 for 10 min

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Table 4 (continued)

Author (Year)	Application	Data	Methods	Results
Zheng H et al. [94]	For short-term traffic flow prediction	PeMS dataset	Attention+ Conv-LSTM + two Bi-LSTM	Best MAPE = 10.1 % for 5 min
Wang et al. [76]	For short-term traffic flow prediction with and without weather factors	Traffic data: open platform openITS + Weather data	Attention mechanism + 1DCNN+LSTM	MSE = 0.00086
Hou Y et al. [26]	Short-term traffic flow prediction with weather conditions	The traffic dataset of metro freeways in the Twin Cities is from the Regional Transportation Management Center + Weather data	Stacked autoencoder (SAE) + RBFNN	Best MAPE = 18.85 %
Lu et al. [39]	Short-term traffic flow prediction	public highway network database published by the British government	ARIMA+LSTM	MAPE = 0.066
Rui Y et al. [62]	Short-term traffic flow prediction	Multiple traffic microwave sensors (RTMS) located at six road sections of Beijing	DBN with ELM+IBF	MAPE = 7.27 %
Li Y et al. [35]	Long-term (next day hourly) traffic flow prediction	Highways from England	W+CNN+LSTM	MAE = 268.359
Mirzahassein H et al. [51]	For short-term traffic volume forecasting	Data from Traffic Control System of Tehran Municipality	WT + GRU +LSTM	Accuracy 94 %
Zhou S et al. [95]	For short-term traffic flow prediction	Highway traffic dataset PeMSD4 from California	Two layer GCN (spatial features) + Bi-LSTM (temporal features) + Attention mechanism	MAPE = 13.95 %
Nigam and Srivastava [54]	For traffic stream variable (flow and speed) prediction	Traffic data from the Caltrans Performance Measurement System (PeMS), Rainfall (weather) data from the National Climate Data Center (NCDC – NOAA)	CNN-LSTM and LSTM-LSTM	Best MAE = 5.87 (5 min) for LSTM-LSTM
Mendez M et al. [48]	Long-term traffic flow forecasting	Four streets of Madrid	CNN+ Bidirectional LSTM	Mean MAE: 82.06 for 12 hour
Wen Y et al. [81]	For traffic flow prediction	PEMS04 and PEMS08 – Performance Measurement System of Caltrans (California Department of Transportation)	RPConvformer	Average MAPE = 13.55 % (PEMS04), 12.44 % (PEMS08)
Li Z et al. [36]	For short-term traffic flow prediction	Flow data from a freeway toll station in Guangdong Province	Self-attention-based hybrid model: encoder-decoder NN + self-attention mechanism	MAPE = 12.99 %
COVID				
Pinter G et al. [58]	to predict the COVID–19 to predict daily infected individuals and mortality rate	COVID–19 cases and mortality rate from Hungary,	hybrid methods of ANFIS and multi-layered perceptron-imperialist competitive algorithm (MLP-ICA)	MAPE = 34.6 % using ANFIS for case prediction MAPE = 33.95 % using ANFIS for mortality rate prediction Hybrid model – MLP-ICA MAPE = 23.15 % for cases prediction MAPE = 36.99 % for mortality rate prediction Forecasting error: 0.02 %
Castillo and Melin [10]	Forecasting confirmed cases and deaths	Humanitarian Data Exchange (HDX)	Hybrid approach: Fractal theory (fractal dimension) and fuzzy logic	Accuracy = 0.86 LSTM AUC score were 62.50 by LSTM for predicting COVID–19 disease MAPE: 1.30 %
Alakus and Turkoglu [3]	prediction of COVID–19 infection	laboratory findings from the Hospital Israelita Albert Einstein at Sao Paulo Brazil	six different model types: ANN, CNN, LSTM, RNN, CNNLSTM, and CNNRNN	
Maleki et al. [44]	To predict the number of confirmed COVID cases and deaths	the China National Health Commission (NHC) and World Health Organization (WHO)	TP-SMN-ARMA model	MSE: 3.76E–05
Hasan N [23]	To predict COVID epidemic	Center for Systems Science and Engineering (CSSE) at the Johns Hopkins University GitHub repository	Hybrid model: EEMD+ANN	
Bhardwaj and Bangia [7]	Estimation of COVID transmission risks	Daily COVID confirmed, deaths and recovered cases data from China, India and USA	Wavelet + Neuronal Fuzzification	Symmetric MAPE for Confirmed COVID cases = 1.73 % in China, 0.15 % in India, 0.16 % in USA MAPE = 1.248
Singh S et al. [67]	Forecast the casualties cases of COVID	COVID–19 deaths in Italy, Spain, France, the United Kingdom, and the United States of America	DWT + ARIMA	The mean SMAPE is 0.25 for the short-term (10 days ahead) and for long-horizon forecasting mean SMAPE is 2.59. MAPE: 4.08 % (CESBAS-ANFIS) MAPE: 4.79 % (Hybrid FPASSA-ANFIS)
Hossein A et al. [25]	Prediction of COVID confirmed cases	Johns Hopkins University's Coronavirus Resource Center,	Hybrid approach: multi-head attention +LSTM + CNN with Bayesian optimization algorithm	
Miodrag Z et al. [50]	To predict new COVID cases	COVID–19 dataset from China	hybridized approach between machine learning, ANFIS and enhanced beetle antennae search swarm intelligence metaheuristics CESBAS-ANFIS method Hybrid FPASSA-ANFIS	MAPE: 3.452 (total cumulative cases) MAPE: 7.140 (Daily new cases) Best MAE = 0056
Prasanth et al. [59]	To forecast the spread of COVID–19 – total cumulative cases	ECDC + Google tend	GWO+LSTM	
Cui S et al. [17]	To predict COVID mortality	COVID data from 79 countries (12th April to 29th July 2020) + climate data	Linear Regression (LR) + SVM + extreme learning machine (ELM)	

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Table 4 (continued)

Author (Year)	Application	Data	Methods	Results
Alvin et al. [4]	Predicting COVID transmission	Twitter data from Kaggle + 10 countries confirmed COVID cases	ODANN: NN + NLP feature extraction	MSE: 0.0000130 (3 days advance). MAE: 0.002972 (3 days advance) – using Skip Gram NLP method
Kumar N et al. [32]	To predict COVID infected cases and deaths	COVID-19 infected cases and deaths in India at the end of month August 2020	ANHFTS + modified Fuzzy c-means (MFCM)	MSE: 191360 for one-month COVID infected cases, 126.22 for one-month COVID death

implemented by integrating the ARIMA + CNN + sequence-to-sequence LSTM models outperformed baseline models (ARIMA, BPNN, LSTM) on S&P 500 index stock dataset for 14 days ahead prediction. Whereas [55], developed a hybrid predictive model called IKN-ConvLSTM which combines six predictors (users' sentiments from tweets, news, forum conversation, Google trends, historical macroeconomic variables, and past stock data) to forecast future stock price movement. Their framework showed an accuracy of 98.31 % in prediction of stock price movement using the Ghana Stock Exchange dataset. In 2022, to increase the accuracy of stock market prediction, [85], developed a three-stage pipeline hybrid model consisting of FastRNN, CNN and Bi-LSTM. Their architecture showed RMSE of 0.205, 0.1513, 0.039 and 0.21 for Apple, Facebook, Nike, and Uber stock values predictions respectively. In order to capture long-term features, short-term features, and features of very long stock prices, [29], implemented a novel hybrid algorithm by combining CNN, LSTM and Attention Mechanism. Their architecture consisted of two arms: Attention-Based CNN (AB-CNN) and Contextual Bidirectional LSTM (CB-LSTM) to do the feature extraction task – output of which concatenated and fed to last stage MLP module. They reported MAPE of 0.921 (on the JNJ dataset), and 0.911 (BAC dataset) in forecasting the stock prices. To overcome the limitation of GAN model in stock market prediction, [70], proposed improvised GAN based Hybrid Prediction Algorithm by combining LSTM, RNN and CNN. Their framework worked with best MAE of 0.0022 for Maruthi stock dataset. [61] first produced a distinctive COVID-19 related PRICE Movement prediction (COVID19 PRIMO) dataset, created incorporating the impacts of social media trends connected to COVID-19 on stock market price movements. Secondly using the PRIMO dataset, they developed a hybrid and parallel deep fusion model for Stock price Movement Prediction (HPSMP). Their model consisted of two parallel routes: (i) CNN with Local/Global Attention modules, and (ii) integrated CNN and Bi-LSTM route. The two parallel routes are connected by a multilayer fusion layer that joins localized features. Their proposed model used blending of information to integrate COVID-19 related data from Twitter with historical data to forecast the market movement and reported that their model can forecast the stock market's variations with more than 66 % accuracy. [30], combined a Bidirectional Cuda DNN LSTM with CNN (BiCuDNNLSTM-1dCNN) to develop a hybrid framework which experimented on five different stock price datasets. Their experiment showed MAE of 0.897 for crude oil, and 0.566 for DAX datasets. Recently in 2023, [88] to investigate the impact of COVID pandemic on stock market, proposed a multi-task learning-based stock market prediction framework called MVCNN-BiLSTM-Att. Their framework comprises three components TCN, and BiLSTM for combining continuous data (market data) with categorical COVID data which are fed into KNN for stock market trend estimation. Their set up performed superior in terms of forecasting the accuracy of the China CSI 300 Index during the COVID 19 time. [93] used emotion/sentiment of users (input to CNN) together with stock price (input to DAE) as predictors to be fed to LSTM for the stock price movement prediction task. Their framework named SA-DLSTM showed MAPE of 1.10 for Hong Kong stock exchange datasets.

Time Series Forecasting in Traffic – Hybrid models for number of vehicles at a junction forecasting/prediction

[94] implemented a novel hybrid model consisting of two modules: Attention based Conv- (to extract spatial features of traffic flow) LSTM

(to extract short-term temporal features) + two Bi-LSTM (to extract the daily and weekly periodic features of traffic flow). They model showed MAPE of 10.1 % for urban traffic flow prediction and 10.8 % for freeway traffic flow prediction. Using traffic flow data and weather conditions as predictors, [77] implemented a hybrid model by combining attention mechanism and 1DCNN-LSTM for a short-term traffic flow prediction. Their experiment showed that adding weather conditions as one of the predictors improved the prediction accuracy of their 1DCNN-LSTM-Attention model. Similarly, [26] examined the disturbance of weather factors in traffic flow by developing a hybrid framework of stacked autoencoder (SAE) and RBF neural network. This model showed MAPE of 10.37 % in traffic flow prediction using weather conditions. Thus, with the data fusion (time periodicity from historical data + disturbance of weather conditions), their proposed model showed improved accuracy in traffic flow prediction. [62] introduced Ensemble Learning Model combined with Improved Bayesian Fusion (IBF) based on the bagging framework developed by Deep Belief Network (DBNs) to make short-term traffic state prediction. Their proposed model demonstrated MAPE of 6.13 % for traffic speed prediction and MAPE of 7.27 % for traffic volume prediction. [35] combined wavelet decomposition, CNN, and LSTM (W-CNN-LSTM) to develop a hybrid model for the prediction of next-day traffic flow. Their model outperformed with MAE of 303.318 compared to other benchmark models. Similar kind of hybrid model, by combining wavelet transform, GRU and LSTM, implemented by [51] to analyze and forecast traffic volume fluctuations which showed MAE of 0.0642. [95] built an architecture consisting of three components: two-layer graph convolutional network (GNU) (to capture the spatial features), and multilayer bi-directional LSTM (to capture the temporal features) together with attention mechanism (to focus on the important information) for traffic flow prediction. Their model predicted with MAPE of 13.95 % during validation using PeMSD4 dataset. Recently, [54] examined the impact of extreme weather conditions like rainfall on traffic stream by implementing two hybrid models: CNN-LSTM and LSTM-LSTM. Among the two, LSTM-LSTM performed better with MAE of 83.2 for long-term Traffic flow and MAE of 13.4 for traffic speed prediction. Similarly for long-term prediction of traffic flow, [48] presented a combination of CNN (to capture hidden valuable features) and Bi-LSTM (to capture temporal features). Their experiment achieved MAE of 146.99 for 72-hours in advance prediction of traffic flow. Recently, [81] proposed an improvised Transformer based model called RPConvformer consisting of causal convolutional sequence embedding layer (with convolutional and causal convolutional) and attention layer with relative position encoding. The original encoding (for extracting historical traffic information) and decoding (for predicting the future traffic state) framework of Transformer is maintained. Their model showed average MAPE of 13.55 % (PEMS04 dataset) and MAPE of 12.44 % (PEMSD8 dataset). [36] implemented self-attention-based hybrid framework by combining encoder-decoder neural network and a self-attention mechanism to predict the short-term traffic flow. To enhance the ability of capturing key information without any loss, they used self-attention mechanism module for feature extraction in their hybrid model. The experiment demonstrated MAPE of 12.99 %.

4.1. Comparison with State-of-the-Art Methods

Our proposed DCent-PredictiveNet is compared with few State-of-

the-art (SOTA) methods such as Autoregressive Convolutional Recurrent Neural Network (ACRNN) [43], DeepAR [65] based on autoregressive recurrent networks, Temporal Fusion Transformer (AutoTFT) [37], Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting [82], Informer [96], Neural Hierarchical Interpolation for Time Series Forecasting (N-HiTS) [11], and Channel-independent Patch Time Series Transformer (PatchTST) [53]. All the SOTA models are implemented and trained using different datasets (Dataset 1: time series data on COVID in Singapore from data world, Dataset 2: Stock prices data from Apple Inc (AAPL) database or Dataset 3: the *Traffic Prediction Dataset* obtained from Kaggle contained the 48120 observations of the number of vehicles each hour in four different junctions) used for our DEEnt-PredictiveNet (Table 5).

It is evident from the Table 5 that our DEEnt-PredictiveNet outperformed compared to the SOTA methods selected in stock price, hourly traffic and daily COVID cases prediction tasks with MAPE of 1.53 %, 11.41 % and 0.03 % respectively. Among DeepAR, ACRNN, AutoTFT and Informer, DeepAR achieved better MAPE of 5.09 % for stock price predictions, N-HiTS, AutoTFT, Informer, Autoformer, PatchTST, ACRNN, and DeepAR, DeepAR performer better with MAPE of 15.57 % for hourly traffic flow prediction, and ACRNN, PatchTST, DeepAR, AutoTFT, and Informer, PatchTST showed best MAPE of 0.04 % for daily COVID cases predictions.

4.2. Significant contribution

The significant contributions of our implemented hybrid framework are,

- Explainable and prediction results are interpretable.
- A feature computation step is powerful due to the integration of both deep and hand-crafted features. The important features responsible for the model's decision are shown (Figs. 5 – 7).
- The implemented hybrid architecture delivers promising results for daily COVID cases prediction with best MAPE of 0.03 %.
- In addition, the proposed hybrid architecture emphasised the evident features (deep and hand-crafted) contributed to the prediction of daily COVID, stock price and hourly traffic thus, exhibiting better performance and making the hybrid model explainable and results interpretable.
- Our model outperformed compared SOTA methods such as ACRNN and DeepAR.
- It is clear from the review of published work and to the best of our knowledge; the proposed explainable hybrid framework DEEnt-PredictiveNet is the first one to be implemented for three different prediction tasks.
- This algorithm is in the process of implemented as a tool for real-time validation.

5. Conclusion

In this work, a novel explainable hybrid framework DEEnt-PredictiveNet which combines both deep and hand-crafted features is proposed for the prediction task. The performance of hybrid architecture implemented is tested using three different datasets, obtained from online open access. The implemented hybrid framework achieved best MAPE of 0.03 % for COVID positive cases prediction, MAPE of 1.53 %, 11.41 % respectively for stock price and hourly traffic prediction. The hybrid framework achieved an outstanding performance for three different applications. In addition, the hybrid framework is interpretable and showed the significant deep and hand-crafted features responsible for the prediction task. The implemented model is showing promising result; thus, further research work is required to enhance the computational efficiency and scalability of our model and to make it more suitable for real-time and multi-terminal deployment scenarios. Even the model can be further improvised and gained deeper insights into the

Table 5

Results of comparison study.

Prediction	Model	MAPE
Closing stock price	DEEnt-PredictiveNet	1.53 %
	ACRNN	99.01 %
	DeepAR	5.09 %
	AutoTFT	54.27 %
	Informer	93.20 %
Hourly Traffic (vehicle numbers) at one junction	DEEnt-PredictiveNet	11.41 %
	NHITS	21.76 %
	AutoTFT	26.56 %
	Informer	48.54 %
	Autoformer	22.68 %
	PatchTST	22.51 %
	ACRNN	31.27 %
	DeepAR	15.57 %
Cumulative daily COVID cases	DEEnt-PredictiveNet	0.03 %
	ACRNN	47.73 %
	DeepAR	0.77 %
	AutoTFT	99.79 %
	Informer	99.99 %
	PatchTST	0.04 %

decision-making processes by (i) applying to few more datasets for prediction task, (ii) combining improved interpretability methods, visualization techniques, and robustness checks.

CRedit authorship contribution statement

Reshma Ramachandra: Validation, Software, Data curation. **Vidya Sudarshan:** Writing – original draft, Supervision, Conceptualization. **Ru San Tan:** Writing – review & editing, Supervision, Conceptualization. **Smit Ojha:** Validation, Software.

Declaration of Competing Interest

We declare there is no conflict of interest in this work.

Data availability

The authors do not have permission to share data.

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